# TASK 5: EXPLORATORY DATA ANALYSIS (EDA)

## Titanic Passenger Dataset

**Objective:** Extract insights using visual and statistical exploration.
**Tools:** Python, Pandas, Matplotlib and Seaborn.

## 1. Dataset Overview

The dataset contains 891 passenger records and 12 columns: survival, passenger class, name, sex, age, family counts, ticket, fare, cabin and embarkation port.

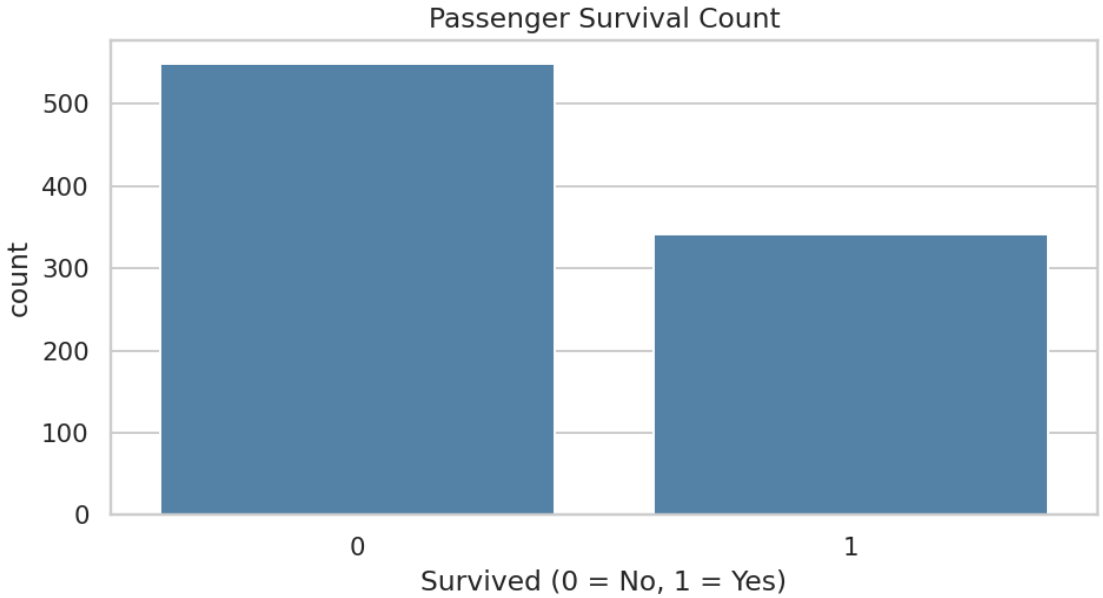
| Measure | Result |
|---|---|
| Rows | 891 |
| Columns | 12 |
| Age non-null | 714 |
| Cabin non-null | 204 |
| Embarked non-null | 889 |
| Survival rate | 38.4% |

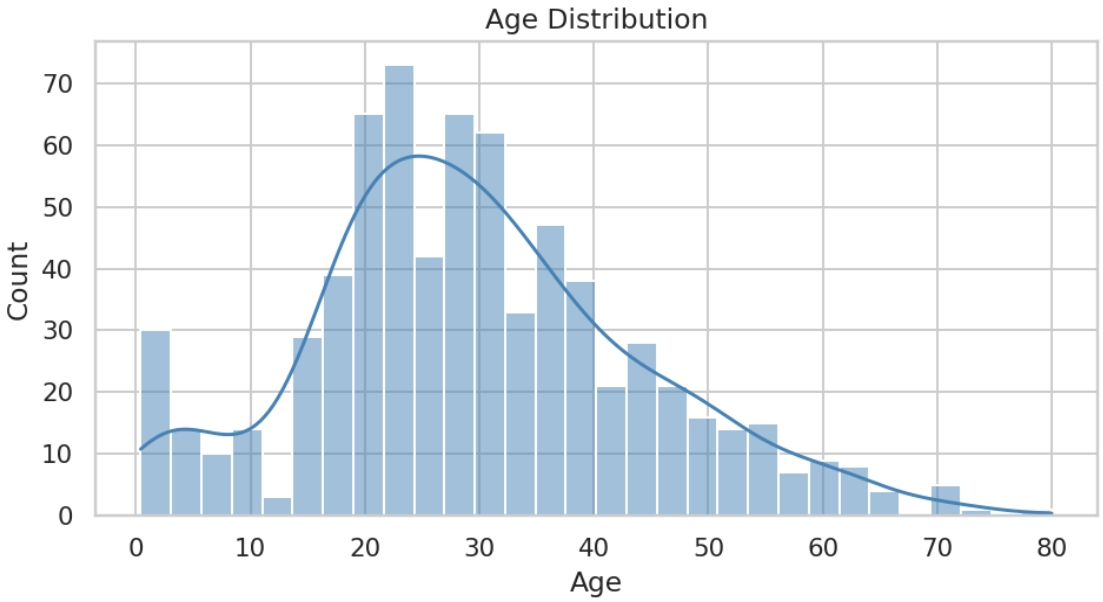
## 2. Data Quality and Preparation

Missing values occur in Age (177), Cabin (687) and Embarked (2). The notebook fills Age with its median and Embarked with its mode in a working copy. Cabin is excluded from selected plots because most values are missing.

## 3. Univariate Analysis

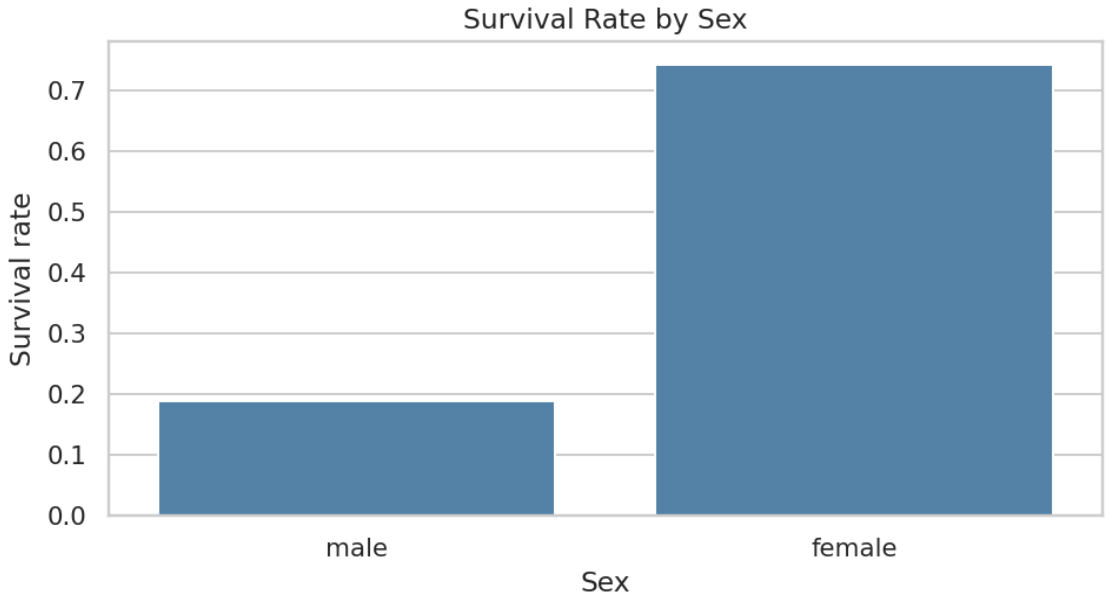
There were 342 survivors and 549 passengers who did not survive. The overall survival rate was 38.4%. The median passenger age was 28 years.

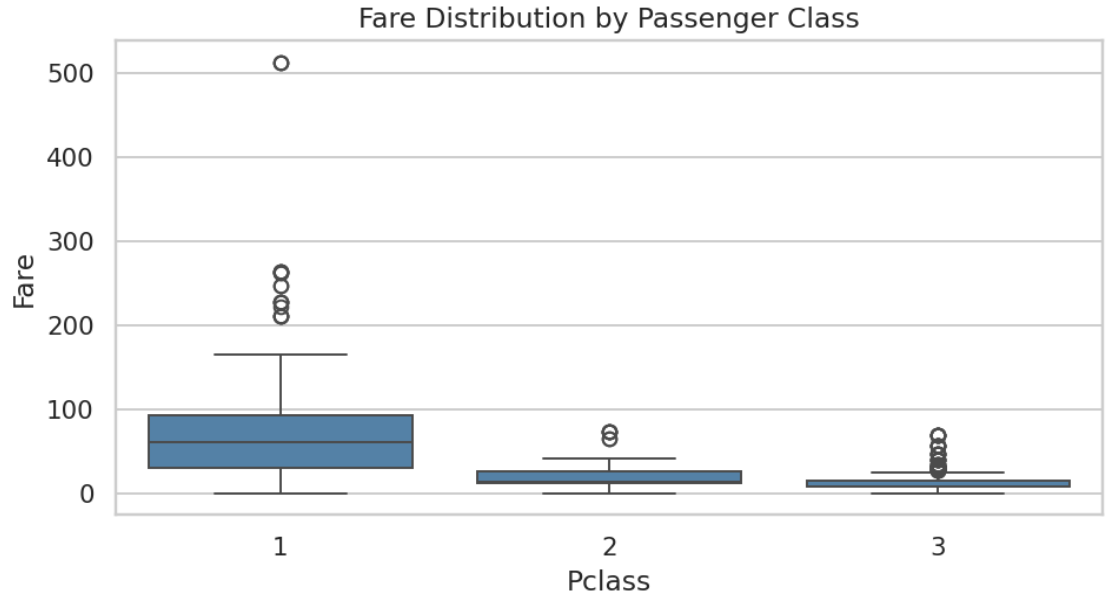



Age Distribution
Count
Age
0
10
20
30
40
50
60
70
80

## 4. Bivariate Analysis

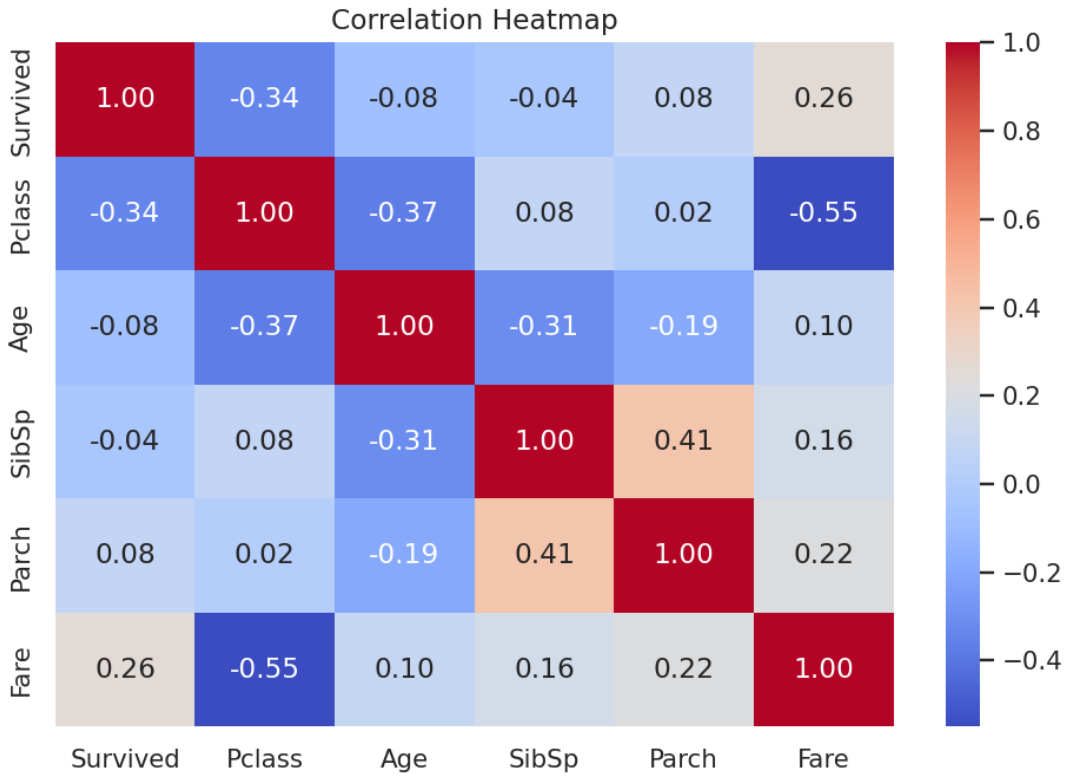
Female passengers had a substantially higher survival rate than male passengers. First-class passengers generally paid higher fares, and the fare boxplot shows high-value outliers.



## 5. Correlation Analysis

The heatmap compares selected numeric variables. Survival is negatively correlated with Pclass (where 1 is first class and 3 is third class) and positively correlated with Fare. Correlation indicates association, not causation.




## 6. Key Findings and Conclusion

- Dataset size: 891 rows and 12 columns.
- Overall survival rate: 38.4%.
- Female passengers had a higher survival rate than male passengers.
- First-class passengers had a higher survival rate than passengers in lower classes.
- Age, Cabin and Embarked contain missing values.
- Fare is widely distributed and includes high-value outliers.

EDA identifies patterns and data-quality issues before further analysis. These findings describe associations in the dataset, not proof of cause and effect.

## 7. Interview Questions and Answers

**1. What is EDA and why is it important?**

EDA explores data through summaries and visualizations to understand structure, quality, patterns and relationships.

**2. Which plots do you use to check correlation?**

A correlation heatmap summarizes numeric correlations; scatterplots show relationships between two numeric variables.

**3. How do you handle skewed data?**

Inspect the distribution, use robust statistics such as the median when suitable, and consider a transformation such as log for positive skew when justified.

**4. How do you detect multicollinearity?**

Use a correlation matrix or heatmap and calculate Variance Inflation Factor (VIF).

**5. What are univariate, bivariate and multivariate analyses?**

Univariate studies one variable, bivariate studies two variables, and multivariate studies three or more variables together.

**6. Difference between heatmap and pairplot?**

A heatmap encodes values in a color matrix; a pairplot displays pairwise plots and distributions.

**7. How do you summarize your insights?**

Report the strongest patterns with supporting numbers and charts, mention limitations, and avoid treating correlation as causation.